Srilakshmi Medarametla

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AI researcher and software engineer focused on evaluation validity, uncertainty, and failure modes in language-model systems, with published work in adversarial NLP and experience building auditable RAG and agentic systems

Research/Publications

RAG Forensics Apr 2026 – Present

- Built an inspectable framework for diagnosing RAG failures by separating retrieval, evidence selection, claim decomposition, and verification rather than collapsing them into a single faithfulness score; evaluated grounding methods on public RAGBench/RAGTruth data using held-out testing and clustered confidence intervals.
- Designed an oracle-evidence intervention to distinguish evidence-selection from verifier failures; replacing similarity-selected evidence with human-annotated evidence reduced false-unsupported judgments from 45.2% to 28.7% on 188 eligible sentences, while residual errors motivated follow-up decomposition experiments

AI Research Investigator May 2023 – May 2024

MIT Lincoln Laboratory *Lexington, MA*

- Alperin, K., Leekha, R., Uchendu, A., Nguyen, T., **Medarametla, S.**, Levya Capote, C., Aycock, S., Dagli, C. *Masks and Mimicry: Strategic Obfuscation and Impersonation Attacks on Authorship Verification.* NLP4DH Workshop @ NAACL 2025
- Red-teamed a BigBird authorship-verification model using lexical substitution, backtranslation, and LLM paraphrasing; evaluated verifier evasion subject to semantic preservation, contributing to the above NAACL paper

AI Training Data Governance — Undergraduate Thesis Aug. 2023 – May 2024

University of Virginia *Charlottesville, VA*

Examined privacy, consent, ownership, and deanonymization risks in AI training data, and how those tensions could shape AI governance

Experience

Software Engineer May 2024 - Present

BTI360 *Herndon, VA*

- Owned development of Python and Java Spring Boot backend microservices and supporting AWS infrastructure spanning databases, search, object storage, and messaging (RDS, DocumentDB, OpenSearch, S3, SQS/SNS)
  - Optimized OpenSearch vector-based RAG kNN by migrating >1M weekly streaming document chunk embeddings from Microsoft Azure-provided OpenAI ADA-002 to TE3L, ensuring backward compatibility and delivering improved performance with ≈ 3.6% decrease in operating costs
  - Enhanced production observability by deploying a Prometheus–Grafana monitoring stack across microservices, adding custom metrics, dashboards, and PromQL alerts, fully managed with Terraform IaC
- Drove architecture and implementation of core capabilities for a domain-specific agentic workflow platform built with React/Next.js/TypeScript: evaluated and selected Temporal for orchestration, Vercel AI SDK for agent definition, MCP for tool integration, and Arize Phoenix/Langfuse for evaluation and observability
- Architected and led technical development of an end-to-end multi-agent pipeline for detecting and contextualizing anomalies in streaming event data, producing auditable intermediate artifacts and evidence-backed outputs

AI Research Investigator June – Aug. 2019,20,21,22

The MITRE Corporation *McLean, VA*

AI-based Knowledge Base T&E with Dr. Flo Reeder: Developed a query-based evaluation procedure for knowledge graphs; simulated degradation by randomly removing relationships at varying rates, measured downstream query performance against a gold standard (observing up to 64% performance degradation), and modeled uncertainty through link prediction and edge-weight perturbations

Education

University of Virginia May 2024

*Bachelor of Science in Computer Science, Bachelor of Arts in Economics – **GPA**: 3.927/4.0* *Charlottesville, VA*

- **Selected coursework**: Artificial Intelligence, Networks, Cybersecurity, Data Privacy, Cryptocurrency, Statistics, Game Theory, Sustainability Economics, Econometrics
- **Activities**: Contrary Capital Venture Partner, HooHacks Sponsorship Committee President, Cavalier Daily Staffer

Thomas Jefferson High School for Science and Technology June 2021

Technical Skills

Python, Java, C/C++, AWS, Docker, Terraform, Temporal, PyTorch, Hugging Face, scikit-learn